



NORTH SYDNEY GIRLS HIGH SCHOOL
HSC Extension 1 Mathematics Assessment Task 1
Term 4, 2012

Name: _____ **Mathematics Class: 11Mx** _____

Time Allowed: 50 minutes + 2 minutes reading time

Total Marks: 47

Instructions:

- Attempt all three questions.
- Start each question in a new booklet. Put your name on every booklet.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Work down the page.
- Do not work in columns.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.

Question	1-3	4-5	6	7	8	Total
PE3	/3		/14		/8	/25
PE4		/2		/14		/16
HE7					/6	/6
	/3	/2	/14	/14	/14	/47

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Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 5.

- 1 Let α , β and γ be the roots of $P(x) = 2x^3 - 5x^2 + 4x - 9$.
Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.
- (A) $\frac{5}{2}$ (B) $\frac{9}{2}$ (C) $\frac{4}{9}$ (D) $\frac{2}{5}$
- 2 Let $P(x) = 2x^3 - x^2 + 2$. Find the remainder when $P(x)$ is divided by $x + 1$.
- (A) -1 (B) 1 (C) 3 (D) 5
- 3 Two roots of $4x^3 + 8x^2 - 9x + k = 0$ are equal in magnitude, but opposite in sign.
Find the value of k .
- (A) 2 (B) -2 (C) -46 (D) -18
- 4 The equation of the chord of contact to the parabola $x^2 = 8y$ from the point $(3, -2)$ is
- (A) $3x - 8y - 8 = 0$ (B) $3x - 4y + 16 = 0$
(C) $3x - 8y + 16 = 0$ (D) $3x - 4y + 8 = 0$
- 5 The equation of the normal to the parabola $x^2 = 4ay$ at the variable point $P(2ap, ap^2)$ is given by $x + py = 2ap + ap^3$.
How many different values of p are there such that the normal passes through the focus of the parabola?
- (A) 0 (B) 1 (C) 2 (D) 3

Section II

42 marks

Attempt Questions 6 – 8

Allow about 42 minutes for this section

Answer each question in a separate writing booklet.

In Questions 6 – 8, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (14 marks) Use a SEPARATE writing booklet

- (a) Find the values of A , B , C and D in the identity 4

$$2x^3 - 3 \equiv (x + 3)(Ax^2 + Bx + C) + D$$

- (b) The polynomial $p(x)$ is given by $p(x) = x^3 - 4x^2 + 4x - 3$

- (i) Show that $x - 3$ is a factor of $p(x)$. 1

- (ii) Express $p(x)$ in the form $(x - 3)(x^2 + bx + c)$, 2
where b and c are integers.

- (c) The cubic equation $3x^3 - x^2 - 38x - 24 = 0$ has roots such that one root is double the reciprocal of a second root.

By letting the roots be α , $\frac{2}{\alpha}$, β and considering the product of all the roots,
solve the equation. 3

- (d) Consider the polynomial $p(x)$, where $p(x) = x^3 - x^2 - 8x + 12$.

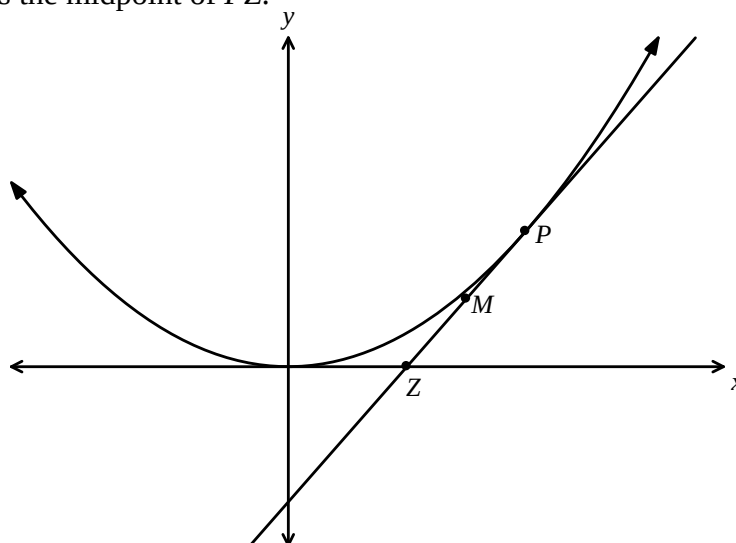
- (i) Factorise $p(x)$ completely. 3

- (ii) The polynomial $q(x)$ has the form $q(x) = p(x)(x + a)$,
where a is a constant chosen so that $q(x) \geq 0$ for all real values of x .
Write down the value of a . 1

Question 7 (14 marks) Use a SEPARATE writing booklet

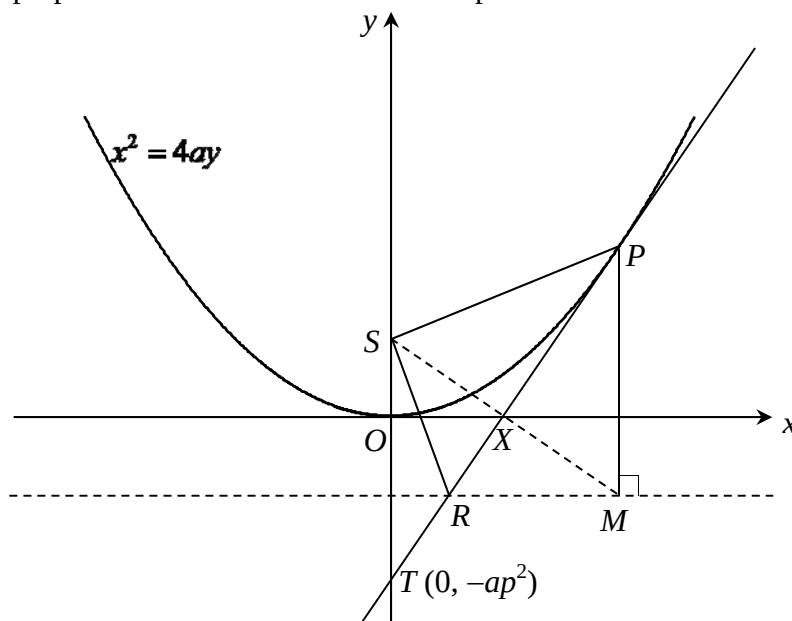
- (a) The tangent at $P(4t, 2t^2)$ to the parabola $x^2 = 8y$ intersects the x -axis at Z .

The point M is the midpoint of PZ .



- (i) Show that the equation of the tangent is $y = tx - 2t^2$. 2
- (ii) Show that the M has coordinates $(3t, t^2)$ 2
- (iii) Find the Cartesian equation of the locus of M as t varies. 2

- (b) In the diagram below, P is the point $(2ap, ap^2)$. PT is the tangent to the parabola at P . PM is perpendicular to the directrix of the parabola.



R and X are the intersections of the tangent with the directrix and x -axis respectively.
 S is the focus.

The equation of the tangent is given by $y = px - ap^2$. (Do NOT prove this)

- (i) Prove that $MPST$ is a rhombus. 3
- (ii) Prove that $\triangle RSP \equiv \triangle RMP$ 3
- (iii) Hence prove that $RMPS$ is a cyclic quadrilateral. 2

Question 8 (14 marks) Use a SEPARATE writing booklet

- (a) Consider the curve $y = \frac{(x-1)^2}{x^2+1}$.
- (i) Show that $\frac{dy}{dx} = \frac{2(x^2-1)}{(x^2+1)^2}$ 2
- (ii) Find the coordinates of the stationary points and determine the nature of the curve's stationary points. 3
- (iii) Write down the equation of the horizontal asymptote. 1
- (iv) Sketch the curve, showing all important features. 2
You are NOT required to find any points of inflexion.
- (b) A curve is defined by the parametric equations $x = t + \frac{1}{t}$ and $y = t^2 + \frac{1}{t^2}$ for $t \neq 0$.
- (i) Find the Cartesian equation of the curve. 2
- (ii) By considering the discriminant, or otherwise, find the values of k for which $x = k$ has solutions, where k is a constant. 2
- (iii) Sketch the curve, showing any domain restrictions implied by the above parts 2

End of paper



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Extension 1 Mathematics Assessment Task 1 Term 4, 2012

Sample Solutions

MC Answers

1. C
2. A
3. D
4. D
5. B

Section I

- 1 Let α, β and γ be the roots of $P(x) = 2x^3 - 5x^2 + 4x - 9$.

Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

- (A) $\frac{5}{2}$ (B) $\frac{9}{2}$ (C) $\frac{4}{9}$ (D) $\frac{2}{5}$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{\frac{4}{2}}{\frac{9}{2}} = \frac{4}{9}$$

- 2 Let $P(x) = 2x^3 - x^2 + 2$. Find the remainder when $P(x)$ is divided by $x + 1$.

- (A) -1 (B) 1 (C) 3 (D) 5

$$P(-1) = 2(-1)^3 - (-1)^2 + 2 = -1$$

So the remainder is -1 .

- 3 Two roots of $4x^3 + 8x^2 - 9x + k = 0$ are equal in magnitude, but opposite in sign. Find the value of k .

- (A) 2 (B) -2 (C) -46 (D) -18

Let the roots be $\alpha, -\alpha$ and β .

$$\therefore \alpha + (-\alpha) + \beta = -\frac{8}{4} = -2$$

$$\therefore \beta = -2$$

$\therefore x = -2$ is a root of the equation

$$\text{Substitute } x = -2 \text{ into the equation: } 4(-2)^3 + 8(-2)^2 - 9(-2) + k = 0$$

$$\therefore -32 + 32 + 18 + k = 0 \Rightarrow k = -18.$$

- 4 The equation of the chord of contact to the parabola $x^2 = 8y$ from the point $(3, -2)$ is

- (A) $3x - 8y - 8 = 0$ (B) $3x - 4y + 16 = 0$
(C) $3x - 8y + 16 = 0$ (D) $3x - 4y + 8 = 0$

The chord of contact is $xx_0 = 2a(y + y_0)$; $a = 2$; $(x_0, y_0) = (3, -2)$.

$$\therefore 3x = 4(y - 2) \text{ is the chord of contact i.e. } 3x - 4y + 8 = 0.$$

- 5 The equation of the normal to the parabola $x^2 = 4ay$ at the variable point

$P(2ap, ap^2)$ is given by $x + py = 2ap + ap^3$.

How many different values of p are there such that the normal passes through the focus of the parabola?

- (A) 0 (B) 1 (C) 2 (D) 3

Only the normal at $(0, 0)$ passes through the focus.

OR

Substitute $(0, a)$ into the equation of the normal i.e. $pa = 2ap + ap^3$

$$\therefore ap^3 + ap = 0 \Rightarrow ap(p^2 + 1) = 0$$

$\therefore p = 0$ i.e. $(0, 0)$ is the only point where the normal passes through the focus.

Section II

Question 6 (14 marks)

- (a) Find the values of A , B , C and D in the identity $2x^3 - 3 \equiv (x + 3)(Ax^2 + Bx + C) + D$ 4

$A = 2$ (comparing coefficients of x^2 .)

Substitute $x = -3 \Rightarrow 2(-3)^3 - 3 = D$

$\therefore D = -57$.

Substitute $x = 0 \Rightarrow -3 = 3C - 57$

$\therefore 3C = 54 \Rightarrow C = 18$.

Substitute $x = 1 \Rightarrow 2 - 3 = 4(2 + B + 18) - 57$

$\therefore B + 20 = 14$

$\therefore B = -6$

$\therefore A = 2, B = -6, C = 18, D = -57$

- (b) The polynomial $p(x)$ is given by $p(x) = x^3 - 4x^2 + 4x - 3$

- (i) Show that $x - 3$ is a factor of $p(x)$. 1

Find $p(3)$: $p(3) = 3^3 - 4 \times 3^2 + 4 \times 3 - 3 = 0$

$\therefore x - 3$ is a factor of $p(x)$.

- (ii) Express $p(x)$ in the form $(x - 3)(x^2 + bx + c)$, 2

where b and c are integers.

From (i): $p(x) = (x - 3)(x^2 + Bx + 1), B \in \mathbb{R}$

Find $p(1)$: $p(1) = 1 - 4 + 4 - 3 = -2$

$\therefore p(1) = -2 = (1 - 3)(2 + B)$

$\therefore 2 + B = 1 \Rightarrow B = -1$

$\therefore p(x) = (x - 3)(x^2 - x + 1)$

- (c) The cubic equation $3x^3 - x^2 - 38x - 24 = 0$ has roots such that one root is double the reciprocal of a second root. By letting the roots be $\alpha, \frac{2}{\alpha}, \beta$ and considering the product of all the roots, solve the equation. 3

$\alpha \times \frac{2}{\alpha} \times \beta = -\frac{-24}{3} = 8$

$\therefore 2\beta = 8 \Rightarrow \beta = 4$

$\therefore x - 4$ is a factor of $3x^3 - x^2 - 38x - 24 = 0$.

$\therefore 3x^3 - x^2 - 38x - 24 = (x - 4)(3x^2 + Bx + 6), B \in \mathbb{R}$

Substitute $x = 1$: $3 - 1 - 38 - 24 = (-3)(9 + B)$

$\therefore 9 + B = 20$

$\therefore B = 11$

$\therefore 3x^3 - x^2 - 38x - 24 = (x - 4)(3x^2 + 11x + 6)$

$\therefore 3x^3 - x^2 - 38x - 24 = (x - 4)(3x + 2)(x + 3)$

$\therefore 3x^3 - x^2 - 38x - 24 = 0 \Rightarrow x = -3, -\frac{2}{3}, 4$

(d) Consider the polynomial $p(x)$, where $p(x) = x^3 - x^2 - 8x + 12$.

(i) Factorise $p(x)$ completely.

3

Test $x = \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

$$p(1) = 1 - 1 - 8 + 12 = 4$$

$$p(-1) = -1 - 1 + 8 + 12 = 18$$

$$p(2) = 8 - 4 - 16 + 12 = 0$$

$\therefore x - 2$ is a factor.

$$\therefore p(x) = x^3 - x^2 - 8x + 12 = (x - 2)(x^2 + Bx - 6), B \in \mathbb{R}.$$

Consider $p(1) = 4$:

$$p(1) = 4 \Rightarrow 4 = (-1)(B - 5)$$

$$\therefore B - 5 = -4 \Rightarrow B = 1$$

$$\begin{aligned}\therefore p(x) &= x^3 - x^2 - 8x + 12 \\ &= (x - 2)(x^2 + x - 6) \\ &= (x - 2)^2(x + 3)\end{aligned}$$

(ii) The polynomial $q(x)$ has the form $q(x) = p(x)(x + a)$,
where a is a constant chosen so that $q(x) \geq 0$ for all real values of x .

Write down the value of a .

1

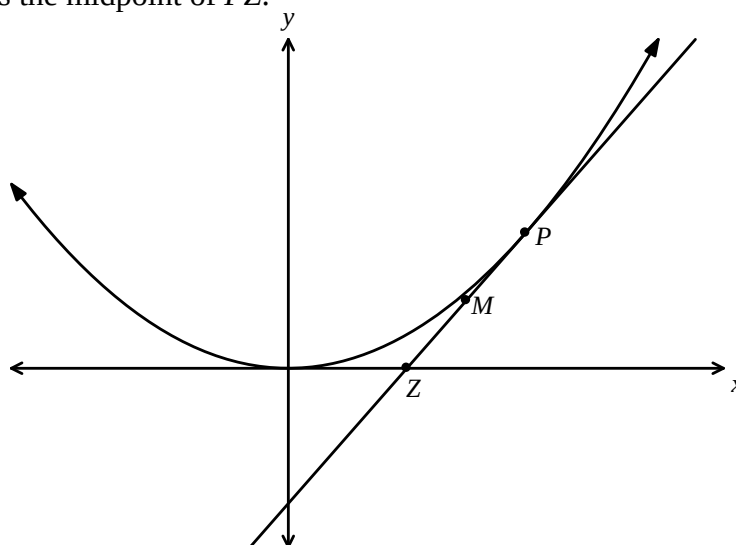
$$q(x) = (x + a)p(x) \geq 0$$

$$\therefore q(x) = (x - 2)^2(x + 3)^2$$

$$\therefore a = 3$$

Question 7 (14 marks)

- (a) The tangent at $P(4t, 2t^2)$ to the parabola $x^2 = 8y$ intersects the x -axis at Z .
The point M is the midpoint of PZ .



- (i) Show that the equation of the tangent is $y = tx - 2t^2$.

2

Parametric	Cartesian
$x = 4t \Rightarrow \frac{dx}{dt} = 4$	$y = \frac{1}{8}x^2$
$y = 2t^2 \Rightarrow \frac{dy}{dt} = 4t$	$\therefore y' = \frac{1}{4}x$
$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4t}{4} = t$	At $P(4t, 2t^2)$, $y' = \frac{1}{4} \times 4t = t$
$\therefore y - 2t^2 = t(x - 4t)$	
$\therefore y = tx - 2t^2$	

- (ii) Show that the M has coordinates $(3t, t^2)$

2

Substitute $y = 0$: Z has coordinates $(2t, 0)$

M is the midpoint of $Z(2t, 0)$ and $P(4t, 2t^2)$.

$$\therefore M \text{ has coordinates } \left(\frac{2t + 4t}{2}, \frac{0 + 2t^2}{2} \right) = (3t, t^2).$$

- (iii) Find the Cartesian equation of the locus of M as t varies.

2

Let $x = 3t$ and $y = t^2$.

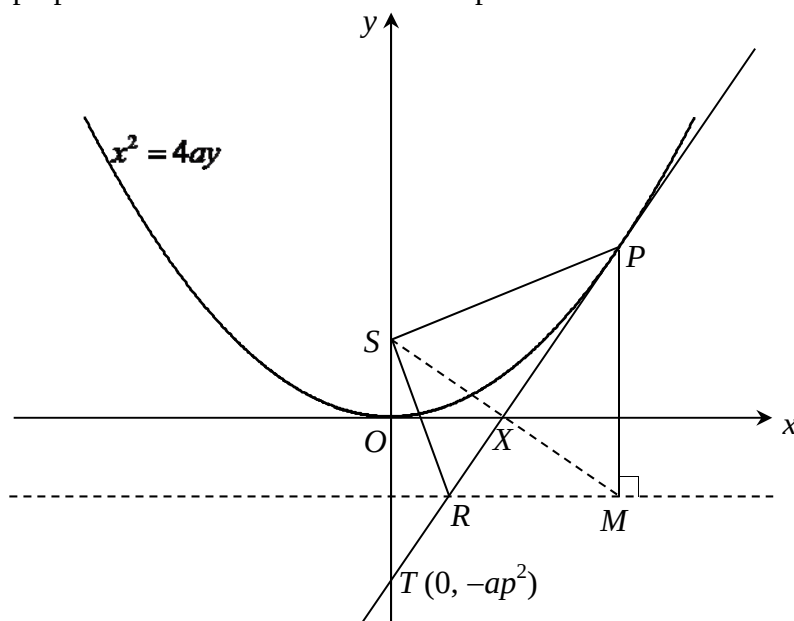
$$x = 3t \Rightarrow t = \frac{x}{3}$$

$$\therefore y = \left(\frac{x}{3} \right)^2$$

$$\therefore x^2 = 9y$$

So the locus of M is $x^2 = 9y$.

- (b) In the diagram below, P is the point $(2ap, ap^2)$. PT is the tangent to the parabola at P . PM is perpendicular to the directrix of the parabola.



R and X are the intersections of the tangent with the directrix and x -axis, respectively. S is the focus.

The equation of the tangent is given by $y = px - ap^2$. (Do NOT prove this)

- (i) Prove that $MPST$ is a rhombus.

3

M has coordinates $(2ap, -a)$.

So using the diagram, $ST = a + ap^2 = PM$.

Using the locus definition of a parabola, $SP = PM$.

$$\therefore ST = PM = SP = a + ap^2.$$

$$\begin{aligned} TM^2 &= (0 - 2ap)^2 + (-ap^2 + a)^2 \\ &= 4a^2 p^2 + a^2 p^4 - 2a^2 p^2 + a^2 \\ &= a^2 p^4 + 2a^2 p^2 + a^2 \\ &= (ap^2 + a)^2 \end{aligned}$$

$$\therefore ST = PM = SP = TM = a + ap^2.$$

Alternative:

Prove that PT & SM are perpendicular

Also prove that SM and PT have the SAME midpoint.

- (ii) Prove that $\triangle RSP \equiv \triangle RMP$

3

$$SP = PM \quad (\text{shown above})$$

$$\angle SPR = \angle MPR \quad (\text{diagonals of rhombus})$$

PR is common.

$$\therefore \triangle RSP \equiv \triangle RMP \quad (\text{SAS}).$$

- (iii) Hence prove that $RMPS$ is a cyclic quadrilateral.

2

$$\angle PSR = \angle PMR \quad (\text{matching sides cong. } \Delta s)$$

$$= 90^\circ \quad (\text{given})$$

Question 8 (14 marks)

(a) Consider the curve $y = \frac{(x-1)^2}{x^2+1}$.

(i) Show that $\frac{dy}{dx} = \frac{2(x^2-1)}{(x^2+1)^2}$

2

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x^2+1) \times 2(x-1) - (x-1)^2 \times 2x}{(x^2+1)^2} = \frac{2(x-1)[(x^2+1) - x(x-1)]}{(x^2+1)^2} \\ &= \frac{2(x-1)(1+x)}{(x^2+1)^2} = \frac{2(x^2-1)}{(x^2+1)^2}\end{aligned}$$

(ii) Find the coordinates of the stationary points and determine the nature of the curve's stationary points.

3

Stationary points when $y' = 0$ i.e. $2(x^2-1) = 0$.

$$\therefore x = \pm 1$$

Stationary points are at (1, 0) and (-1, 2)

NB Test only the numerator as $(x^2+1)^2 > 0$

x	-2	-1	0	1	2
y'	6	0	-2	0	6
	/	-	\	-	/

(1, 0) is a rel. min. turning point and (-1, 2) is a rel. max. turning point.

(iii) Write down the equation of the horizontal asymptote.

1

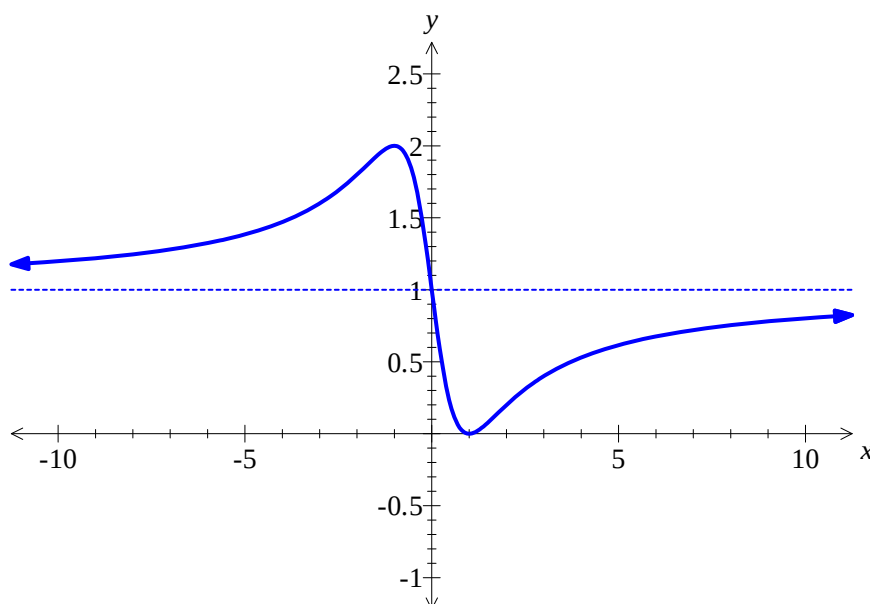
$y = 1$. Why? $\lim_{x \rightarrow \infty} \frac{(x-1)^2}{x^2+1} = 1$

(iv) Sketch the curve, showing all important features.

2

You are NOT required to find any points of inflexion.

x-int: $x = 0$; y-int: $y = 1$; No symmetry



(b) A curve is defined by the parametric equations $x = t + \frac{1}{t}$ and $y = t^2 + \frac{1}{t^2}$ for $t \neq 0$.

(i) Find the Cartesian equation of the curve.

2

$$\begin{aligned} y &= t^2 + \frac{1}{t^2} \\ &= \left(t + \frac{1}{t}\right)^2 - 2 \\ &= x^2 - 2 \end{aligned}$$

(ii) By considering the discriminant, or otherwise, find the values of k for which $x = k$ has solutions, where k is a constant.

2

Considering the discriminant

$$x = t + \frac{1}{t} = k$$

$$\therefore t^2 - tk + 1 = 0$$

$$\therefore \Delta = k^2 - 4$$

To have solutions, $\Delta \geq 0$.

$$\therefore k^2 - 4 \geq 0$$

$$\therefore (k - 2)(k + 2) \geq 0$$

$$\therefore k \leq -2, k \geq 2$$

Or otherwise

For $t > 0$, $t + \frac{1}{t} \geq 2$ and so if $t + \frac{1}{t} = k$ then $k \geq 2$.

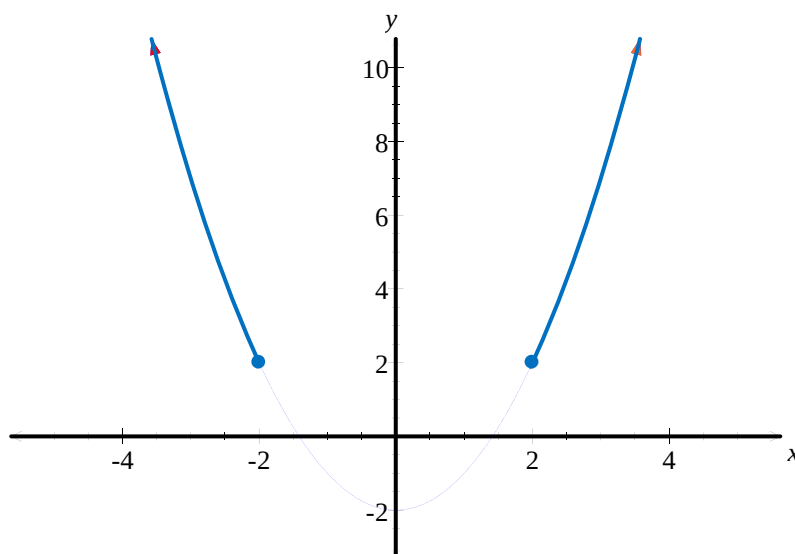
Similarly, if $t < 0$ then $t + \frac{1}{t} \leq -2$ and so $k \leq -2$

$$\therefore k \leq -2, k \geq 2$$

(iii) Sketch the curve, showing any domain restrictions implied by the above parts

2

Part (ii) gives the domain restrictions on the locus i.e. $x \leq -2$, $x \geq 2$ as the values of k are the possible values that x can take.



End of solutions